

CHUMPURUK, Russian Federation

WMO: 245380

Lat: 64.233N

Lon: 116.917E

Elev: 456

StdP: 14.46

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-63.1	-58.8	-65.2	0.1	-63.8	-61.3	0.1	-58.5	10.0	-33.2	9.5	-32.1	3.3	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	84.0	65.0	80.1	63.2	76.1	60.8	67.6	79.4	64.9	76.5	62.4	73.5	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	88.4	73.6	60.2	79.3	69.8	57.8	72.5	67.2	32.4	79.6	30.2	76.7	28.4	73.8	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.4	9.8	DB	-63.0	89.5	5.0	3.6	-66.6	92.1	-69.5	94.2	-72.3	96.2	-76.0	98.8	(4)
(5)			WB	-63.0	71.4	5.0	4.8	-66.6	74.9	-69.5	77.7	-72.3	80.4	-76.0	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	-36.0	-25.7	-2.2	19.4	38.9	56.6	61.1	55.4	40.6	18.0	-17.0	-31.5		
	DBStd	36.81	15.33	15.25	15.46	12.26	9.69	8.36	7.32	6.89	7.85	12.80	14.43	14.87		
	HDD50	13578	2666	2121	1618	917	365	30	4	29	298	992	2011	2527		
	HDD65	18318	3131	2541	2083	1367	808	274	166	305	733	1457	2461	2992		
	CDD50	809	0	0	0	0	22	227	349	196	15	0	0	0		
	CDD65	74	0	0	0	0	0	21	46	7	0	0	0	0		
	CDH74	1154	0	0	0	0	16	382	613	139	4	0	0	0		
	CDH80	305	0	0	0	0	3	103	179	20	0	0	0	0		
	WSAvg	3.6	2.5	2.9	4.1	4.9	4.7	4.3	3.8	3.6	3.7	3.5	2.8	2.8		
	PrecAvg	10.7	0.4	0.3	0.3	0.5	0.8	2.0	1.4	1.5	1.1	1.1	0.7	0.5		
	PrecMax	14.0	0.9	1.1	1.1	1.0	1.4	4.5	3.6	3.9	2.4	2.4	1.5	0.8		
(15) Precipitation	PrecMin	8.0	0.2	0.1	0.0	0.0	0.2	0.5	0.3	0.0	0.1	0.2	0.2	0.1		
	PrecStd	1.9	0.2	0.2	0.2	0.3	0.4	1.1	0.7	0.9	0.5	0.6	0.3	0.2		
	DB	14.4	17.0	42.2	52.6	75.2	88.5	89.4	83.0	71.6	45.3	26.8	14.7			
	MCWB	13.2	14.6	34.2	40.9	54.6	65.6	68.7	63.7	56.3	38.5	24.8	13.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	5.6	11.0	34.6	46.4	68.3	83.9	85.0	78.0	63.4	40.4	18.1	7.9			
	2% MCWB	4.9	9.9	29.7	37.0	50.5	62.8	68.1	62.3	52.5	36.8	17.3	7.1			
	5% DB	-4.6	5.0	29.6	42.7	62.7	79.5	81.3	73.9	57.8	37.1	11.5	0.5			
	5% MCWB	-5.1	4.2	25.9	34.9	48.1	60.9	64.9	60.3	49.2	34.1	10.6	-0.1			
	10% DB	-12.9	-1.7	23.6	39.4	56.9	75.1	77.1	69.8	53.4	34.2	5.0	-7.5			
	10% MCWB	-13.3	-2.5	20.6	32.6	44.9	58.8	63.1	58.6	46.5	31.6	4.4	-7.8			
	0.4% WB	13.0	14.7	34.6	41.5	55.9	67.9	73.8	66.2	58.1	40.0	24.9	13.5			
	0.4% MCDB	14.0	17.0	40.7	50.7	73.5	83.3	85.9	78.6	70.6	43.1	26.9	14.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	5.0	10.0	29.7	37.6	51.5	65.1	70.0	63.8	53.9	36.9	17.2	7.2			
	2% MCDB	6.0	11.0	34.0	44.6	64.8	80.2	82.1	74.1	60.4	39.6	18.2	8.0			
	5% WB	-5.1	4.1	25.9	35.6	48.7	62.5	66.9	61.7	50.6	34.4	10.8	-0.1			
	5% MCDB	-4.6	4.8	29.6	42.0	61.3	75.7	77.1	71.3	57.0	37.0	11.8	0.4			
	10% WB	-13.2	-2.4	20.7	33.2	45.7	60.1	64.2	59.6	47.3	31.8	4.4	-7.8			
	10% MCDB	-12.9	-1.7	23.7	38.9	56.0	72.3	74.9	67.7	52.4	34.0	5.0	-7.3			
	MDBR	15.9	23.1	29.9	26.7	20.0	22.1	21.6	21.3	17.9	16.2	18.0	15.8			
	5% DB	23.8	28.0	27.8	27.5	30.1	31.3	28.2	29.9	26.0	14.9	22.2	26.0			
(36) Mean Daily Temperature Range	5% WB	23.3	27.0	23.0	19.2	16.0	13.0	13.2	15.4	15.0	11.9	21.8	25.4			
	MCDBR	23.4	27.4	27.9	24.1	27.5	26.4	24.9	26.5	24.8	14.0	22.4	26.4			
	MCWBR	23.1	26.4	23.3	17.5	15.2	12.2	13.6	14.6	15.5	11.6	22.0	25.7			
	taub	0.202	0.253	0.294	0.329	0.357	0.404	0.452	0.453	0.321	0.269	0.194	0.190			
(40) Clear-Sky Solar Irradiance	taud	1.887	2.054	2.069	2.100	2.309	2.300	2.140	2.069	2.468	2.303	1.913	1.878			
	Ebn at Noon	108	206	246	265	270	257	237	221	244	205	108	40			
	Edh at Noon	9	20	31	39	35	36	41	41	22	16	8	3			
	RadAvg	62	340	878	1521	1772	1815	1681	1218	736	332	103	23			
(44) All-Sky Solar Radiation	RadStd	4	19	31	78	72	144	96	99	78	32	9	1			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (1 neighbor)		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	-360	N/A	N/A	

Nomenclature: See separate page